

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

April 2023

21BT305 - UNIT OPERATIONS IN BIOPROCESS INDUSTRIES**★ 17BT302 - UNIT OPERATIONS**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss about different types of fluids and their classification.	10	L*2	1	1
b)	A simple U-tube manometer containing mercury is connected to a pipe in which a fluid of specific gravity 0.8 having vacuum pressure is flowing. The other end of the manometer is open to atmosphere. Find the vacuum pressure in pipe, if the difference in mercury levels in the two limbs is 40 cm and the height of the fluid in the left from the center of pipe is 15 cm below.	8	L3	1	2
c)	When is Rayleigh's method of dimensional analysis preferred?	2	L1	1	1
2. a)	Differentiate between the following:				
i)	Steady and unsteady flow	5	L1	2	1
ii)	Uniform and non-uniform flow				
b)	A 25 cm diameter pipe carries oil of specific gravity 0.9 at a velocity of 3 m/s. At another section, the diameter is 20 cm. Find the velocity at this section and also the mass flowrate of oil.	5	L2	2	2
c)	Derive an expression for head loss due to sudden expansion in the cross section of a pipe.	10	L3	2	2
3. a)	Using Buckingham's pi-theorem show that the velocity through a circular orifice is given by $v = \sqrt{2gH} \phi [D/H, \mu/\rho vH]$, where H is the head causing flow, D is the diameter of the orifice, μ is the coefficient of orifice, ρ is density and g is acceleration due to gravity.	10	L3	1	2
b)	Derive Bernoulli's equation of motion.	10	L3	2	2
Unit – II					
4. a)	State the assumptions of Kynch theory of sedimentation.	5	L1	3	1
b)	Explain about the construction of plate and frame filter press.	5	L2	3	1
c)	When is centrifugation employed? Explain about any one type of centrifugation equipment.	10	L2	3	2
5. a)	Explain about turbines and paddles agitators.	10	L2	4	1
b)	Why is size reduction important? What are the different mechanisms of size reduction?	10	L2	4	1
6. a)	Derive the expression for terminal settling velocity in Stoke's region.	10	L3	3	2
b)	A turbine agitator is installed in a tank. The turbine is operated at 90 rpm and the liquid in the tank has a viscosity of 10 cP and a density of 929 kg/m ³ . Calculate the required power of the mixer. The turbine diameter is 0.61 m. (Data: For turbine agitator, N_p value from the power curve in laminar region is 65 and in turbulent region is 5.75).	10	L3	4	2
Unit – III					
7. a)	Derive the flow equation for an orificemeter.	10	L3	5	2
b)	How does a centrifugal pump work? Explain.	10	L2	5	1
8. a)	What is a rotameter used for? With a neat sketch, explain its working principle.	10	L2	5	1
b)	A horizontal venture meter with inlet and throat diameters 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 20 cm of Hg. Determine the rate of flow. Take C_d as 0.98.	10	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
